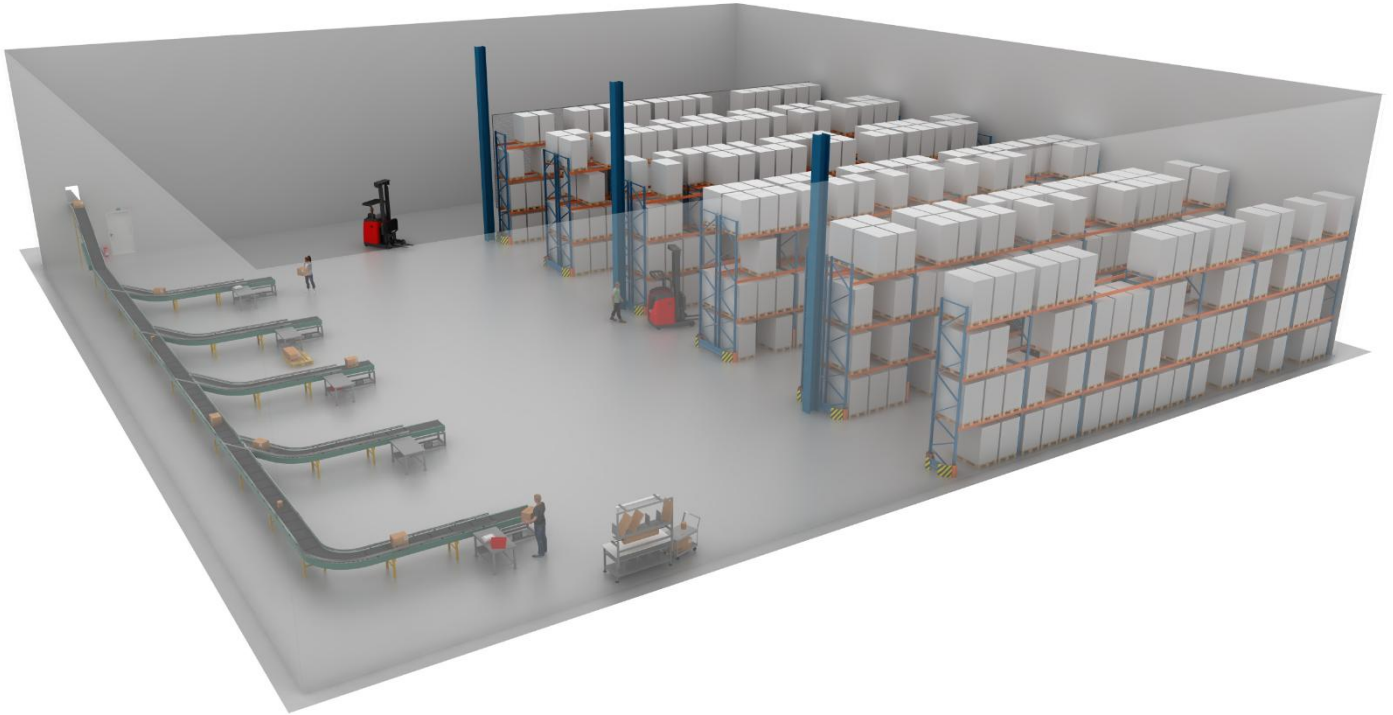




Manuals CSA

Enhanced Client Presentation

This manual explains a number of features within CET that can be used to provide an enhanced project presentation for clients. This manual will explain how to create a building, how to add accessories to the building and which render settings may be useful.

5-11-2025

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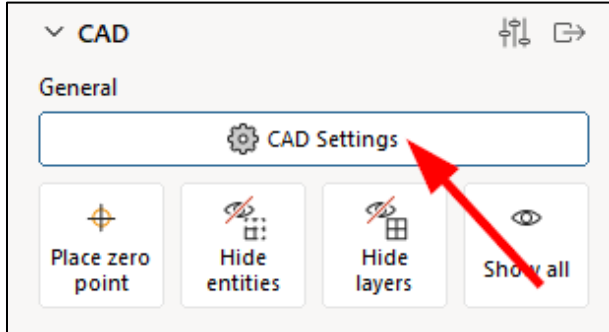
Contents

1. Adding a floor plan in CSA	2
DWG file.....	2
PDF file.....	4
2. Drawing walls on the floor plan.....	5
In case of a flat roof.....	5
In case of an angled roof	7
Columns.....	11
3. Structural parts.....	12
Steel beams	12
Horizontal beams (angled beams).....	13
Vertical beams (columns)	14
Trusses	15
4. Lights	17
Placing lights inside the building	17
Groups	18
Light settings.....	19
5. Accessories	20
Materials.....	20
Doors	21
Other accessories	24
6. Rendering	27
Photograph setup settings.....	28
Post processing setting	31
7. Extra tips.....	34
Advanced selection.....	34
Alternatives	35
Copying objects	36
Filters	37
Filter: 2D and 3D view	37
Filter: Paper view.....	38
Appendix A: Effect of adding accessories to a project.....	39
Appendix B: Effect of color temperature.....	41

1. Adding a floor plan in CSA

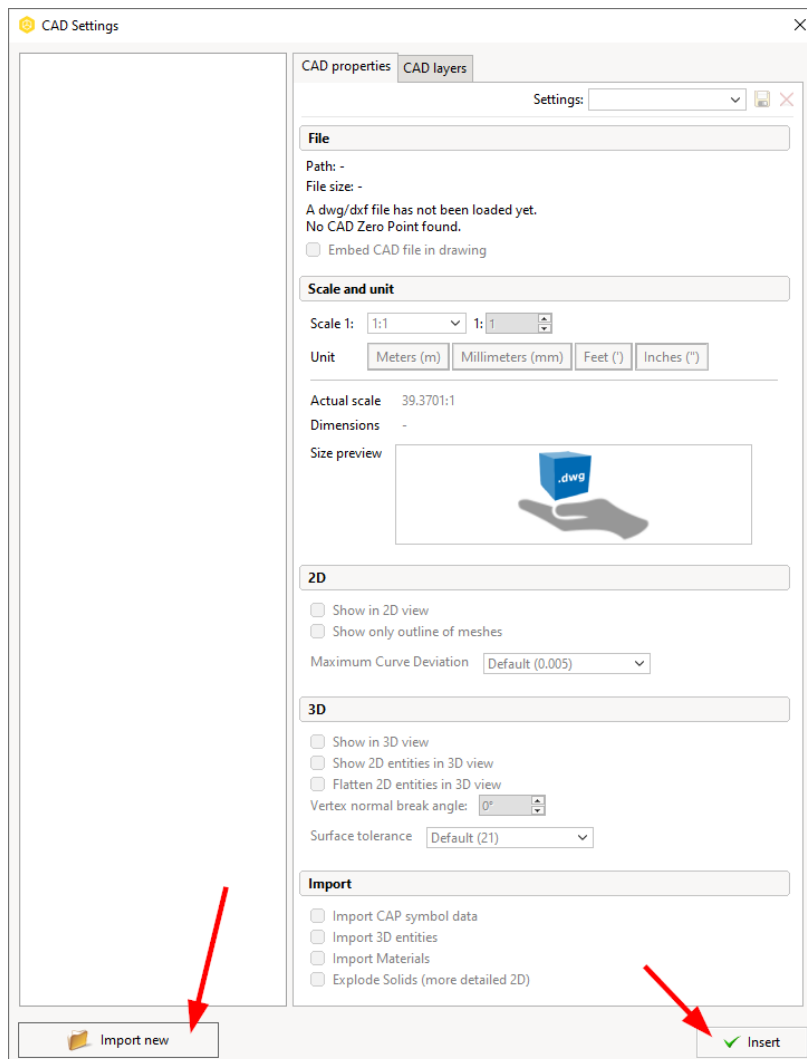
DWG file

When you are in the 2d view, look under the “Tools” tab on the left side of your screen, here you’ll find “CAD”. Click on the “CAD Settings” icon.

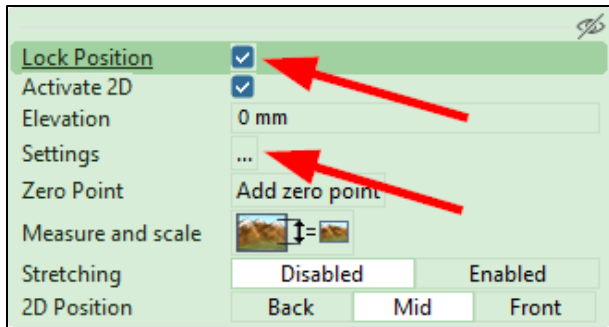


The following screen will open. You can import your drawing under “Import new”. Browse for your DWG file.

Click on “insert” to insert the layout into CET.

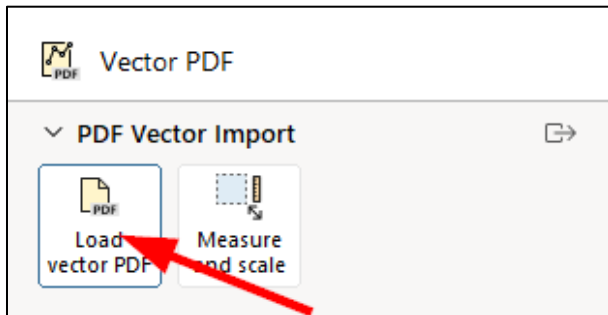


After placing the layout, you can lock the position of the layout. First select the layout to open the green box and tick the “Lock Position” checkbox in the green box. This will prevent that the layout can accidentally be moved or deleted. Click on the three points next to “Settings” to open the “CAD Settings” screen again. In this screen it is possible to change several settings for the layout.



PDF file

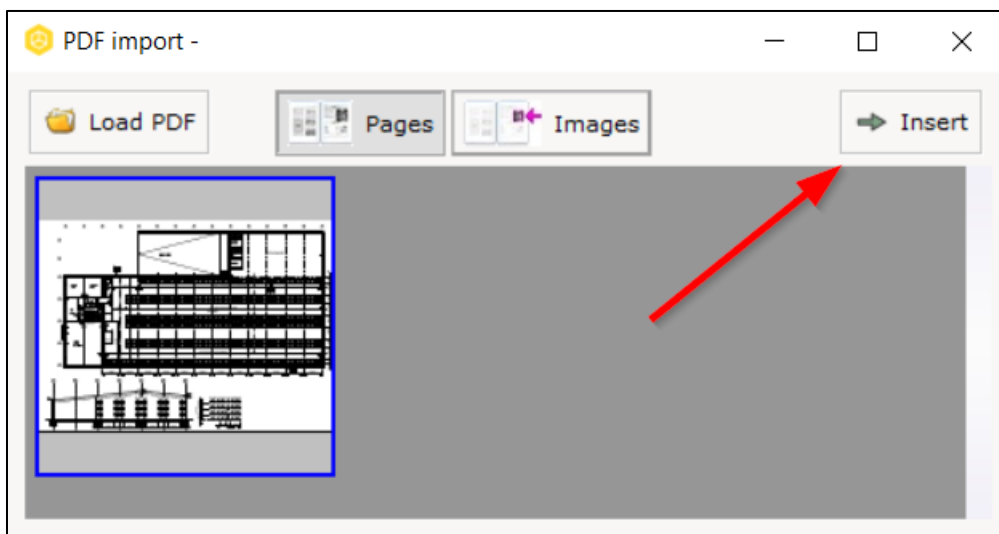
When you are in the 2D view, go to the “PDF” tab, click on “Load vector PDF”.



Insert the rectangle on your screen by clicking on your screen to place the first corner of the rectangle and click a second time to place the opposite corner.

Double click on the added rectangle, now you can browse and select the PDF file you want to add as a floor plan to the drawing.

A new tab will open, click on the added PDF-file to select it. After the file is selected, there will be a blue box around it. Click on “Insert” to insert the PDF-file.



Now your PDF-file has been added.

Go to the “PDF” tab, click on “Measure and Scale PDF” to scale the layout.

Select two points on the PDF whose distance is known.

Enter the actual length in the green box that appears.

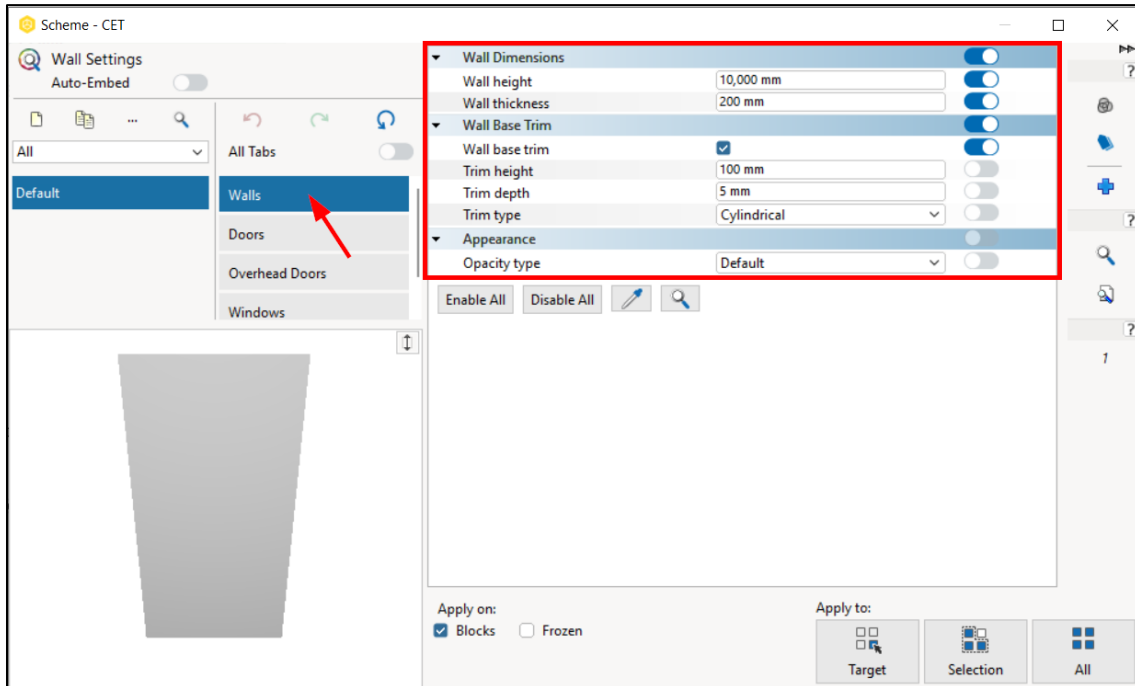
You may want to check other dimensions if they are approximately correct, as PDF files have lower accuracy than DWG files.

2. Drawing walls on the floor plan

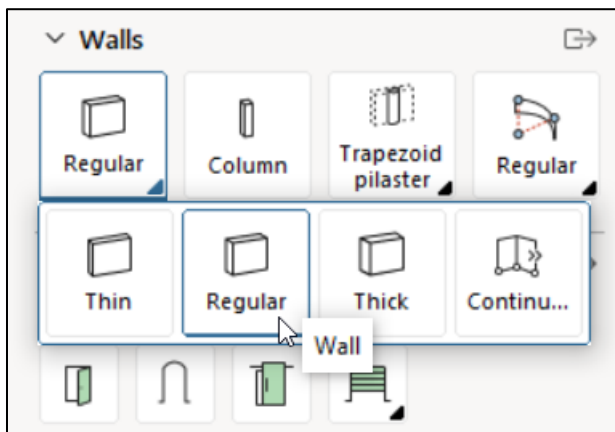
This chapter explains how to draw walls on the floor plan.

In case of a flat roof

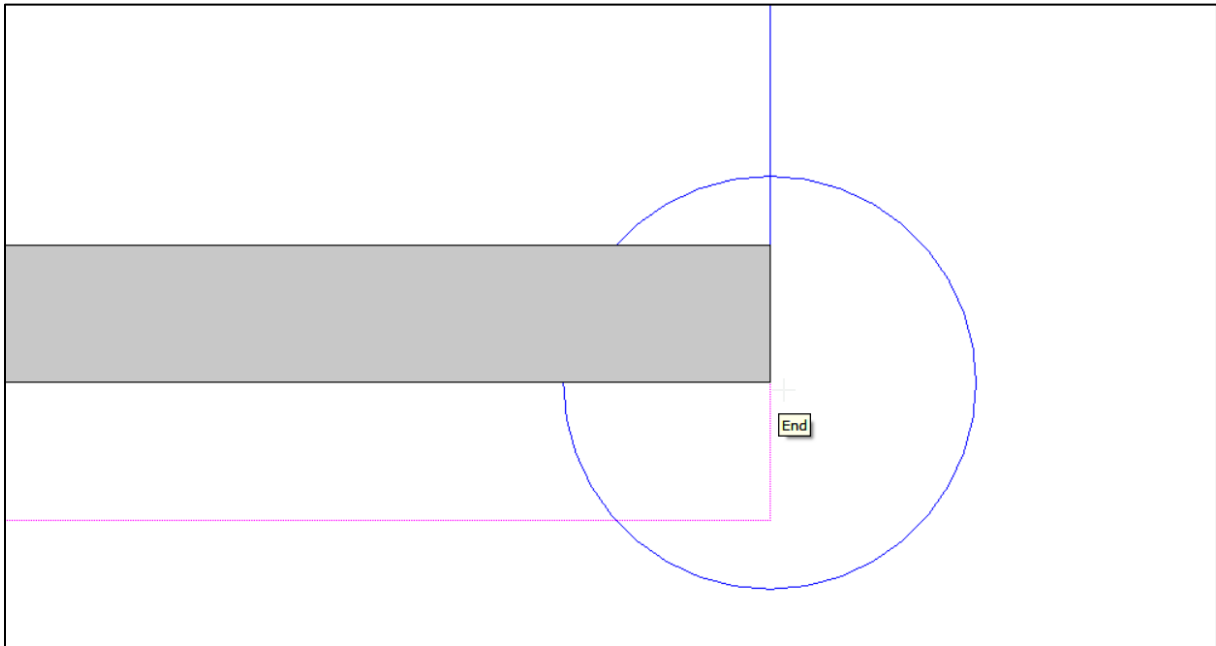
First go to the “Wall Drawing” tab, click on “Wall scheme”  to set the height and thickness of the wall.



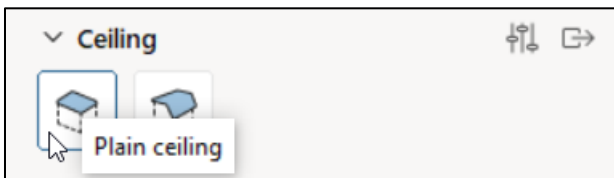
Now you can add the walls to your drawing, the easiest method is to do this in the 2D view. Under “Walls” click on the “Regular” icon to place a wall with the chosen settings.



Now you can click on the starting point of your wall, it is easiest to choose a corner of a wall on your layout. Here you will see a text like “end” or “intersection”, this are places where the starting point of your wall will snap to. Now you can draw a straight wall to the next corner, place it by clicking the end or intersection line on your layout. After you have placed the wall, it is possible to change on which side of your layout line the wall is, by hovering your mouse to the correct side and clicking again. You can also change the side of your wall in the green box before you place the wall by checking or unchecking the “right” checkbox. The remaining walls can be placed in the same way, it is also possible to snap the wall to an already existing wall.



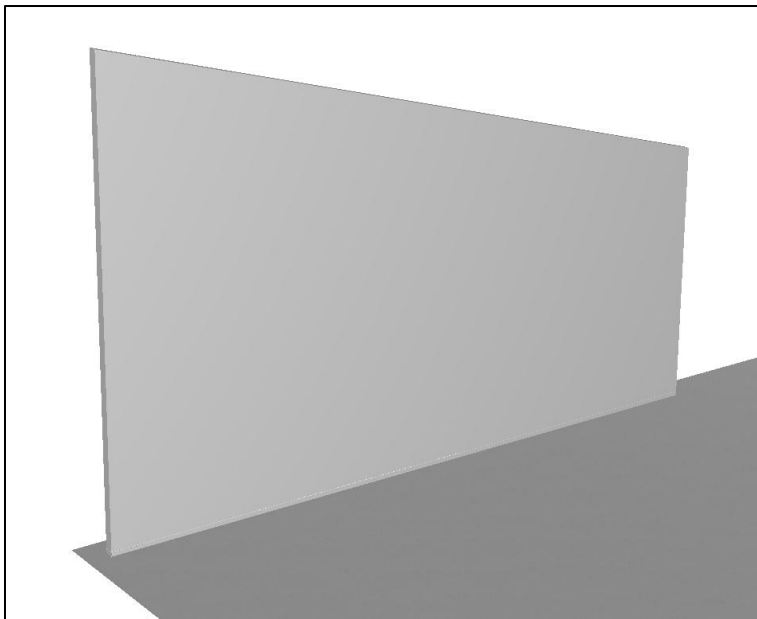
After placing all the surrounding walls, it is time to place the ceiling/roof in the model. It is only possible to place rectangular shaped ceilings, you can place multiple rectangular ceilings to make the complete ceiling. Underneath “Ceiling” in the “Wall” tab, click on “Plain ceiling” to begin placing the ceiling. Click on three opposite corners on the outer wall to place the ceiling, make sure the ceiling height is the same as the height of the walls. For rendering it is also important that the ceiling is completely closed, so no outside light can enter the building through the ceiling.



In case of an angled roof

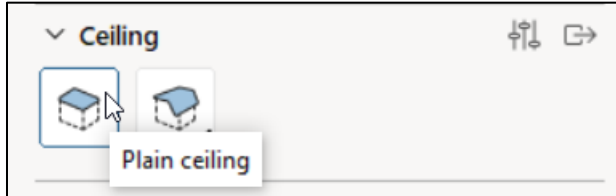
Start by clicking on “Regular” underneath “Walls” in the “Wall drawing” tab to start placing walls. Before you click on a corner to place your wall, make sure that wall height settings for that corner are correct. Change the wall height in the green box when necessary. Now you can click on the starting point of your wall, it is easiest to choose a corner of a wall on your layout. Before you place the opposite side of your wall, change the wall height in the green box. After this you can select another corner or line to place the opposite end of the wall. It is also possible to change the height of the wall after placing it, by selecting the wall and changing the height in the green box.

Length	35,000 mm
Height	10,000 mm
Left height	10,000 mm
Right height	10,000 mm
Elevation	0 mm
Thickness	200 mm
Wall base trim	☒
Trim type	Cylindrical
Trim height	100 mm
Trim depth	5 mm
Opacity type	Default
Hatch Style	Default
More Properties	

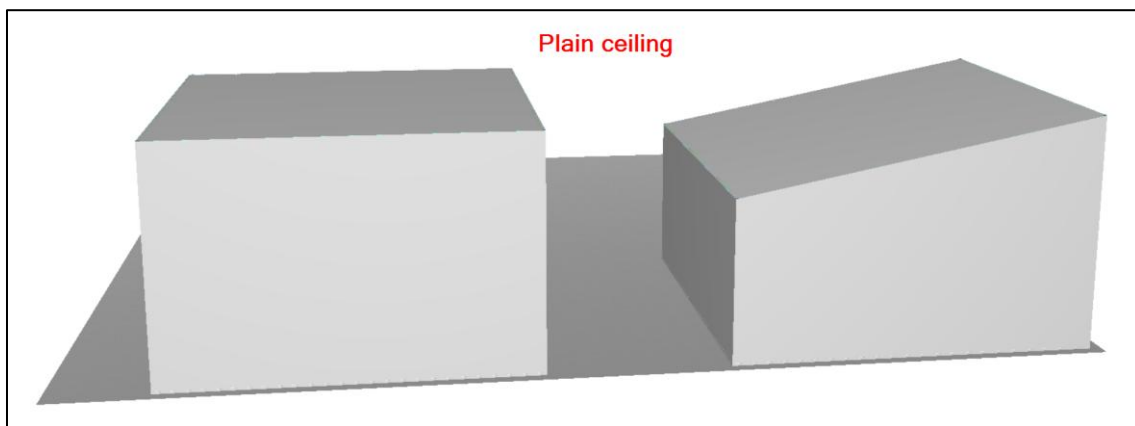


There are 3 different kinds of ceilings to select in the “Wall drawing” tab, these can be found underneath “Ceiling”. These ceilings will be explained below.

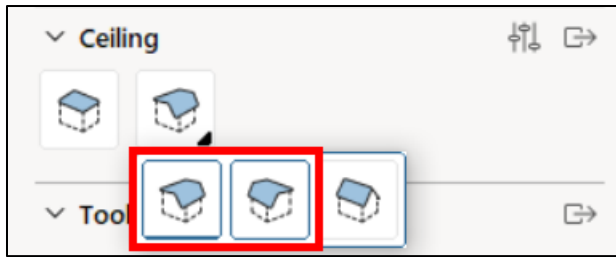
Plain ceiling



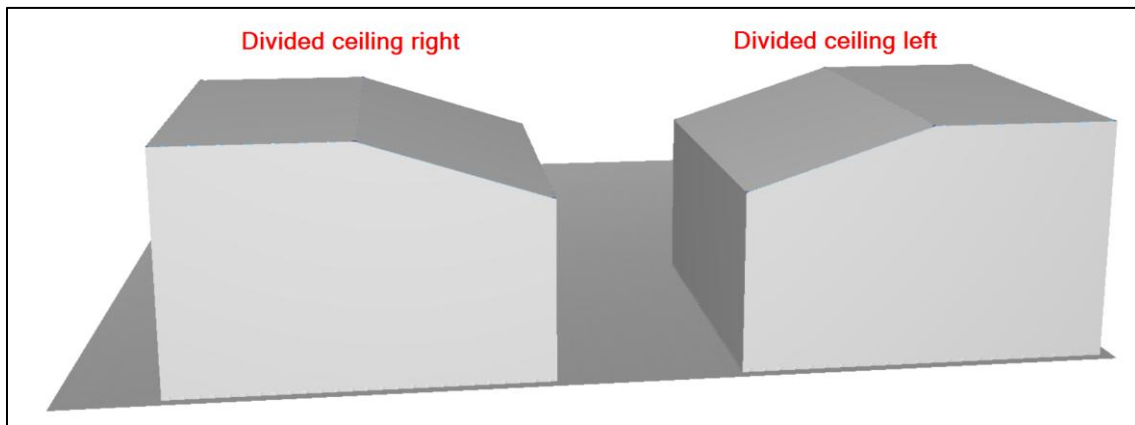
Plain ceilings can be used for flat horizontal ceilings or angled ceilings. How to place a horizontal flat ceiling is described in chapter [Walls in case of a flat roof](#). You can place a plain ceiling on an angle by setting the height of the first corner in the green box and clicking on a corner. Set the correct height of the other end of the ceiling in the green box and click on the other end of the ceiling. Click for a third time on the opposite end to set the length of the ceiling. Both examples can be seen in the example below.



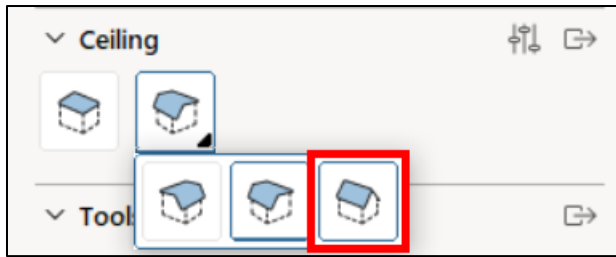
Divided ceiling right/left



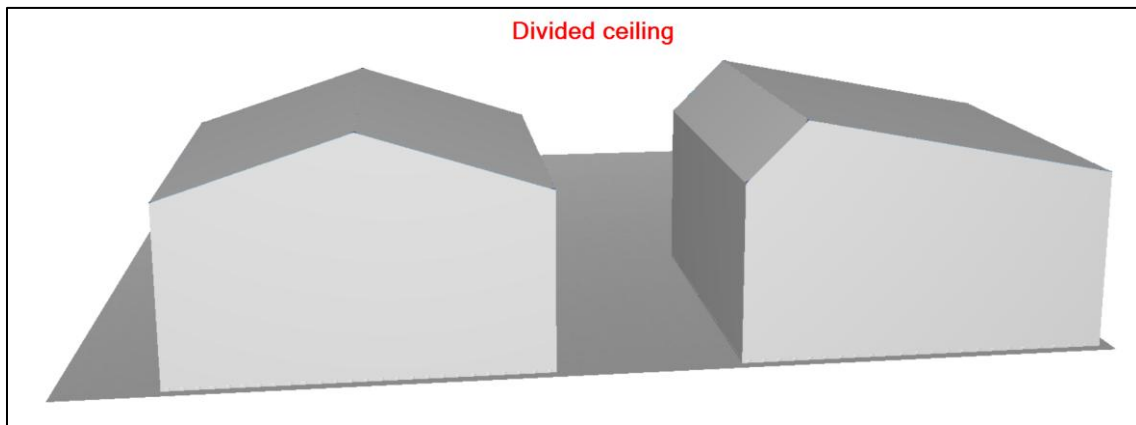
Use the “Divided ceiling right/left” when you want a part of the ceiling to be angled and the other part horizontal, both examples can be seen below. This ceiling type can be placed the same way as the plain ceiling. After placing the ceiling, it is possible to change the heights in the green box. The position of the transition from angled to flat ceiling can be modified by selecting the ceiling and moving the yellow square on the transition line.



Divided ceiling



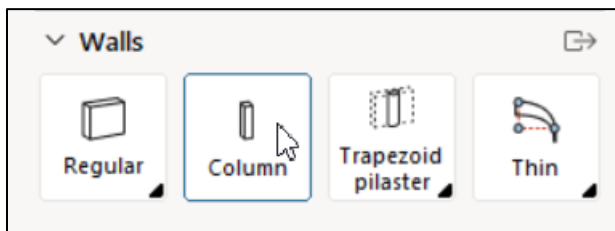
Use the “Divided ceiling” to create a ceiling with two angled sides. You can place this ceiling type the same way as the plain ceiling. After placing the ceiling, it is possible to change the mid height in the green box. If the highest part of the ceiling is not in the middle, it is possible to change the position by dragging the yellow square in the middle of the ceiling to the correct position. See the example below.



Columns

First go to the “Wall Drawing” tab, click on “Wall scheme”  to set the desired thickness and height of the column. Both settings can be changed under “Wall dimensions”, the columns use the same settings as the walls. In case the columns need to be as high as the ceiling, choose a height that is higher than the roof. The roof will cut the column automatically. When you want the columns to have a different material than the walls, it is easier to change the material before placing the columns. How to change the material can be seen in chapter [materials](#).

Now you can add the columns to your drawing, the easiest method is to do this in the 2D view. Under “straight Walls” click on the “Column” icon to place a column with the chosen settings. When needed, helping lines can be used to help positioning the columns.



3. Structural parts

This chapter explains how to place structural parts in the building.

Steel beams

First go to the “Structural” tab, this subchapter explains how to place horizontal, vertical and angled beams. In the image below, the different functions are numbered. Each number will be explained below.

1. First you can select which standard beams you want to use. There is an option for standard AISC beams, which stand for American Institute of Steel construction. This chapter will explain how to set up the Custom steel beams.
2. Here you can choose which Beam type you want to use.
3. Here you can change the dimensions of the chosen beam type. Some parameters only become available when a certain "beam type" is selected.
4. This will let you set the orientation of the chosen beam. When an angled beam is desired, choose the Horizontal orientation.
5. Use this button to start placing the beam in your project. The next part will explain this process further.
6. Here you see the recently used beams. When you want to use beam with the same settings as an already existing beam in your project, you can select the beam here.

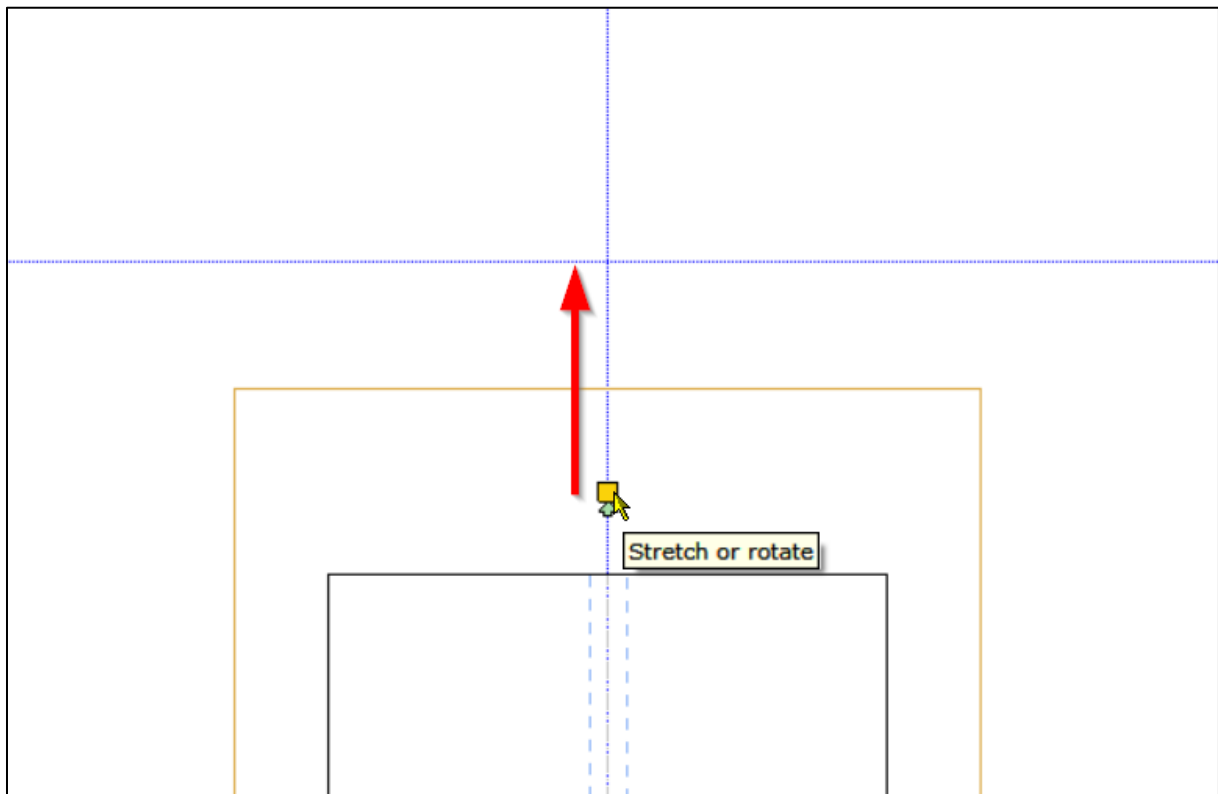
The image shows a software interface for configuring steel beams. The 'Structural' tab is active. Under 'Steel Beams', the 'Custom' option is selected (1). The 'Beam Type' is set to 'Channel' (2). A red box highlights the dimension fields: 'Diameter' (127 mm), 'Width' (127 mm), 'Height' (229 mm), 'Thickness' (25 mm), and 'Web Thickness' (25 mm) (3). The 'Orientation' is set to 'Horizontal' (4). Below these is a 'Preview' section showing a blue C-channel. At the bottom, there is an 'Insert Beam' button (5), 'Mass insert' and 'Replace beam profiles' buttons, and a 'Recently used' section (6).

Horizontal beams (angled beams)

First you need to select a starting point, the easiest way is to do this in the 2D view. It can help to place helping lines in your drawing. After selecting the starting point, click on the end point of the beam. After placing the beam in the drawing, check the height in the 3D view. The height is determined from the place the "beam" is snapped to. The height can be changed after the beam is placed in the green box. "Elevation 1" changes the height of the starting point of the beam, "Elevation 2" changes the height of the end point of the beam. Be aware that when you change the value for "Elevation 1", the value for "Elevation 2" changes by the same amount.

Roll	0°
Elevation 1	0 mm
Elevation 2	0 mm
Beam Type	I-Beam 0.3 x 0.3 x 0.025 x 0.02
Width	300 mm
Height	300 mm
Thickness	25 mm
Web Thickness	20 mm
Length	24,000 mm

Changing the elevation of both side of the beam makes it possible to put the beams on an angle. First change "Elevation 1" to the desired value, then change "Elevation 2". The beam will be angled, but the length of the beam has not changed. Therefore, both ends of the beam need to be stretched to the correct snapping point. Click on the yellow square and drag it to the correct position. After placing a beam, it is often desired to copy the beam over the length of the building so all the beams have the same settings. This process will be explained in chapter [Copying objects](#).



Vertical beams (columns)

Vertical beams are suitable to be used as columns. First you need to select a starting point, the easiest way is to do this in the 2D view. It can help to place helping lines in your drawing. Before selecting the starting point, the height(length) can be changed in the green box.

Orientation	Horizontal	Vertical
Base Point	P0	P1 P2 P3 C
Elevation 1	0 mm	
Length	9,144 mm	
Rotation	0°	

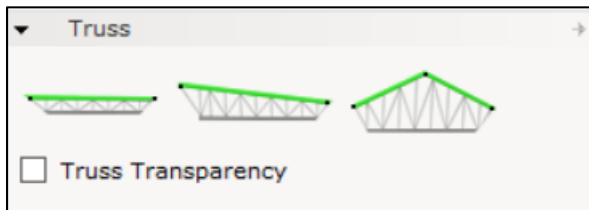
After the vertical beam has been placed, the height(length) can still be changed. You can do this by selecting the beam and changing the Length in the green box.

Elevation 1	0 mm
Beam Type	I-Beam 0.3 x 0.3 x 0.025 x 0.02
Width	300 mm
Height	300 mm
Thickness	25 mm
Web Thickness	20 mm
Length	9,144 mm

Look in chapter [Copying objects](#) for an explanation on how to copy vertical beams.

Trusses

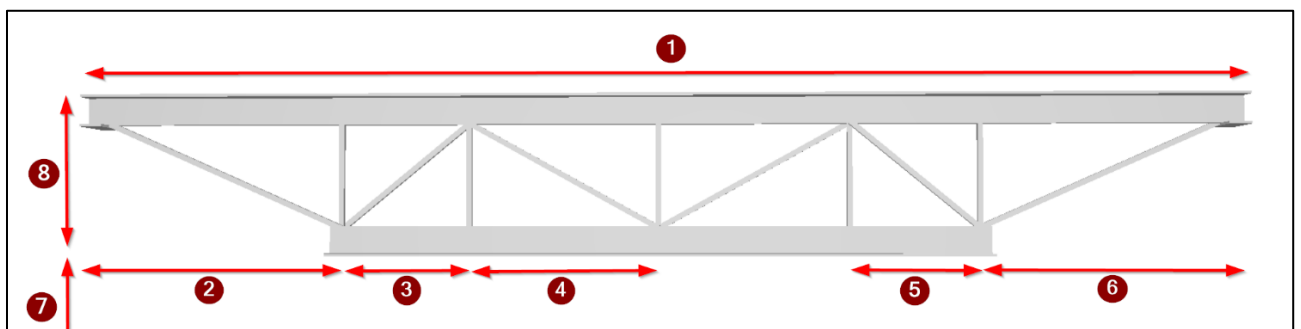
In the “Structural” tab it is also possible to place trusses. Choose one of the three truss options. In the pictures you will see the differences between the truss options



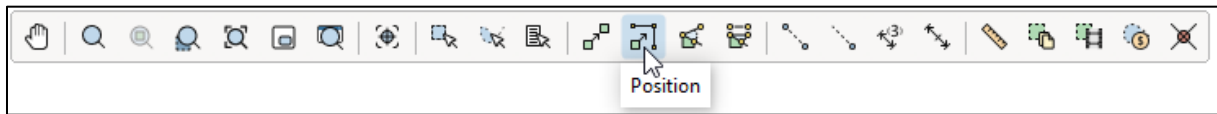
Now click (preferably in the 2D view) close to the desired position to place the truss in the drawing. After placing the truss in the drawing, click on the truss to open the green box and change the settings of the truss. In the image below, the different options are numbered. Each number will be explained below. The same numbers are also shown in the image of the truss.

1. This is the total length of the truss. This value depends on the values at options 2 to 6. The length can only be adjusted in steps equal to the value in option 4. Change the length at the end, after step 2 to 6 are completed.
2. Distance between left side top beam to left side bottom beam.
3. Spacing between the two vertical struts on the left side.
4. Spacing between two vertical struts on the inside part of the truss. How many sections there are depends on the total length of the beam.
5. Spacing between the two vertical struts on the right side.
6. Distance between Right side top beam to Right side bottom beam.
7. Distance from floor to bottom truss.
8. Height of the truss from bottom to top. When the angled or the triangle truss are chosen, other options are shown to adjust the height of the truss on both ends points. For the triangle truss also the Mid Point Height can be adjusted.

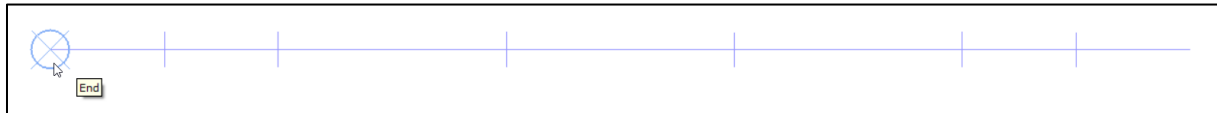
Length	10,058 mm	1
Panel Point (PP) 1	1,219 mm	2
PP 2	610 mm	3
Intermediate Spacing	914 mm	4
PP 2 from End	610 mm	5
PP 1 from End	1,219 mm	6
Elevation	9,144 mm	7
Height	762 mm	8



After dimensioning the truss, it is time to put the beam in the correct position. When in the 2D view, click on **Position** in the toolbar.



Click on a point on the truss where you want the truss to position from, an end point for example. Click on another point on your drawing or building where you want to move the truss to.



Look in chapter [Copying objects](#) for an explanation on how to copy trusses.

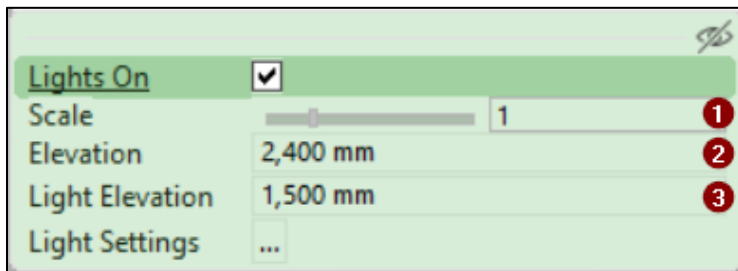
4. Lights

Lights are necessary when you want to make renders of your model. In the basic CET, there are 4 lights to choose from. Download the **Lights Extension** from the CET Marketplace when you want more light sources to pick from. This manual will explain how to use the lights from this extension because these lights have more settings available.

Placing lights inside the building

First go to the “Lights” tab. This chapter will explain the use of “Ceiling Lights” in particular. Select a light from the extension and place it in the desired location by clicking, preferably in the 2D view. Click on the light after placing it to change the settings in the green box. For this example, the “Cone light” is used. In the image below, the different settings are numbered. Not all settings will be available with all “Ceiling Lights”. Each number will be explained below.

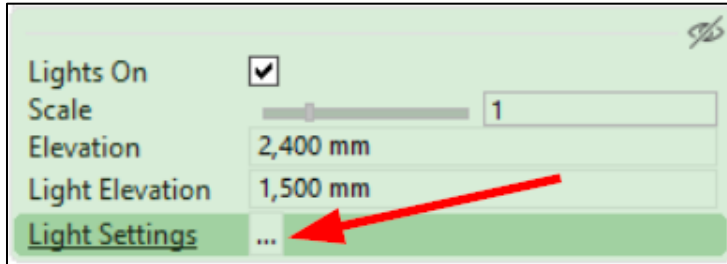
1. Light size scale, use the slider or type a number to scale the size of the light.
2. The height of the top of the light compared to the floor. When the light is attached on a cord, this setting changes the height of the top of the cord.
3. This setting is only available with lights attached to a cord. This is the distance from the floor to the bottom of the light.



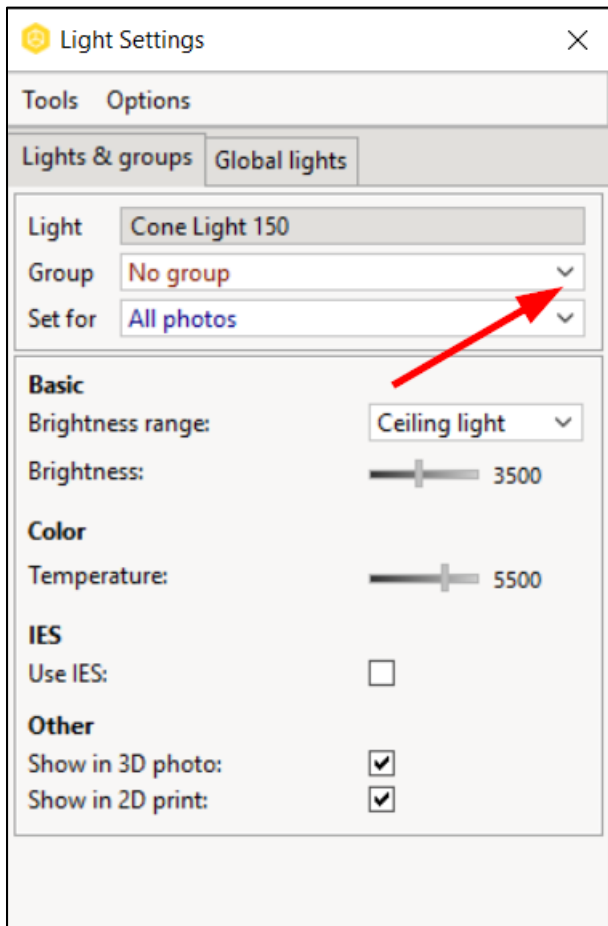
Look in chapter [Copying objects](#) for an explanation on how to copy lights. When copying a light, the copied lights will have the same settings.

Groups

Groups can be used to change the Light Settings for multiple lights simultaneously. When a light from a group is copied, the copied light will also be assigned to the same group. To assign a light or multiple lights to a group, first select the light(s). Click on the three dots in the green box to open “Light Settings”.

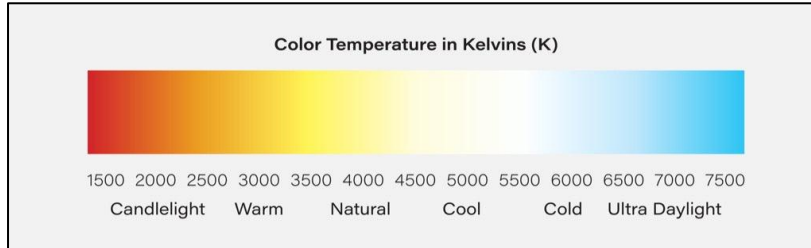


In the Light Settings, click on the arrow next to Group. Here you can select to which group the lights need to be assigned to or you can create a new group.



Light settings

1. Choose a type of light here, this affects the brightness you can choose at the second setting. Choose "Floodlight" to make sure you have enough indoor light for rendering, this is needed for big open spaces.
2. The brightness level of the lights, the value to choose from is determined by the first setting. Choose the highest value possible to make sure you have enough indoor light for rendering.
3. The slider controls the color temperature of the lights in Kelvins from 1500K to 7500K. See [Appendix B](#) for examples which effects the color temperature has on renders.



4. Selecting this checkbox lets you pick a different IES profile than the default used for the light. IES profiles control how the light is being spread.

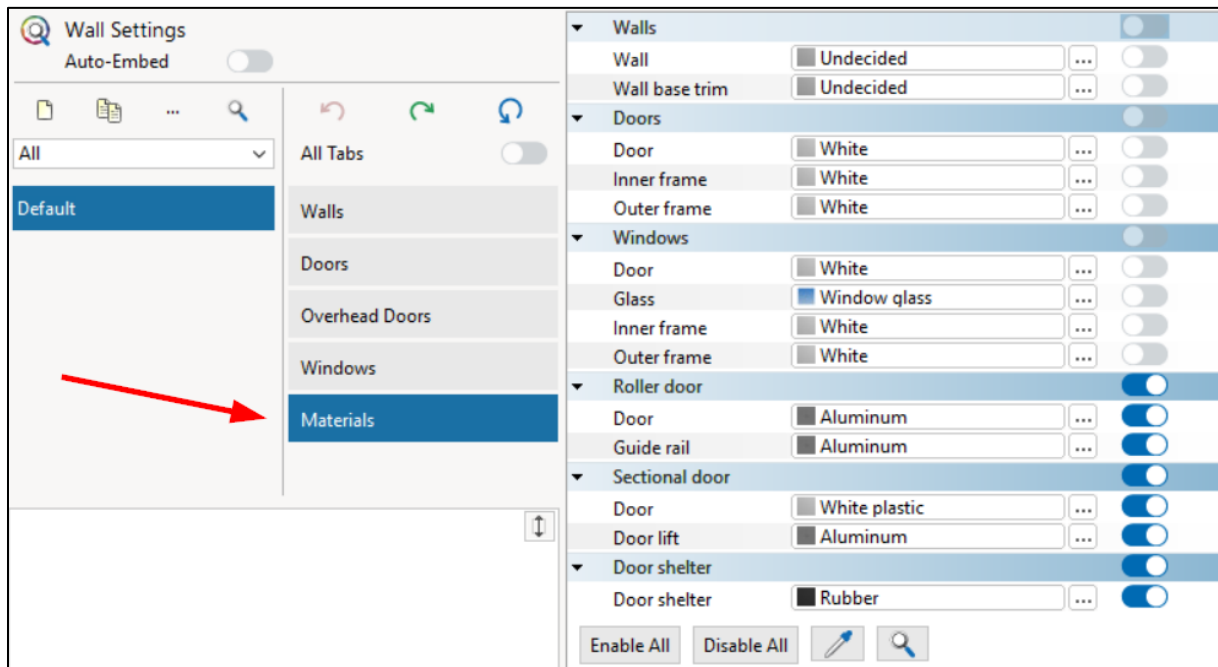
The image shows a "Light Settings" dialog box with a close button (X) in the top right corner. The dialog is divided into two tabs: "Tools" and "Options", with "Options" currently selected. Under the "Options" tab, there are two sub-tabs: "Lights & groups" and "Global lights", with "Lights & groups" selected. The "Lights & groups" sub-tab contains three dropdown menus: "Light" (set to "Cone Light 150"), "Group" (set to "No group"), and "Set for" (set to "All photos"). Below these are three sections: "Basic", "Color", and "IES". The "Basic" section has "Brightness range:" (set to "Ceiling light") and "Brightness:" (a slider set to 3500). The "Color" section has "Temperature:" (a slider set to 5500). The "IES" section has "Use IES:" (a checkbox that is unchecked). Below these are two checkboxes under the "Other" section: "Show in 3D photo:" (checked) and "Show in 2D print:" (checked). Red circular numbers 1 through 4 are overlaid on the image to highlight specific settings: 1 points to the "Brightness range:" dropdown, 2 points to the "Brightness:" slider, 3 points to the "Temperature:" slider, and 4 points to the "Use IES:" checkbox.

5. Accessories

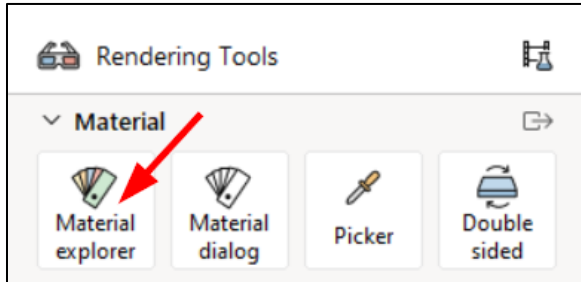
This chapter discusses accessories that can improve the rendering model further. There will be an explanation for assigning materials to the building or to other objects, but also accessories that can be added to the building. Through the CET Marketplace it is possible to download more materials and accessories.

Materials

For changing the materials of the wall and accessories placed in/on the wall, go to the “Wall Drawing” tab and click on “Wall settings”. Go to “Materials” as can be seen in the picture below. On the right side, materials can be selected by clicking on the box where the material or color is displayed. This will open a new screen where the materials can be selected. For the best results in rendering, select a PBR material. PBR stands for “Physically Based Rendering”, these materials are textured and more realistic.

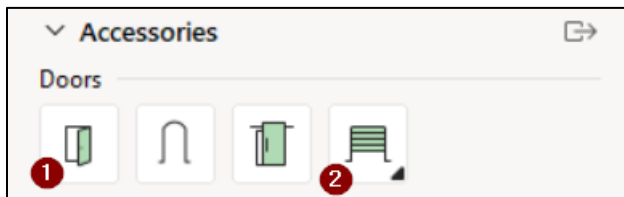


It is also possible to assign materials to objects/buildings through “Material explorer”, you can find this in the “Rendering Tools” tab. This will open a new screen where the materials can be selected. After selecting a material move the cursor to the model, a paint bucket icon will show up next to the cursor. Hover with the cursor on an object/building/floor, a box can be seen around the object to show which part the material will be assigned to. For some objects it is also possible to only assign a material to a part of the object. This can be seen in the green box, change it by pressing the **tab** button. Click the left mouse button to confirm changing the material.

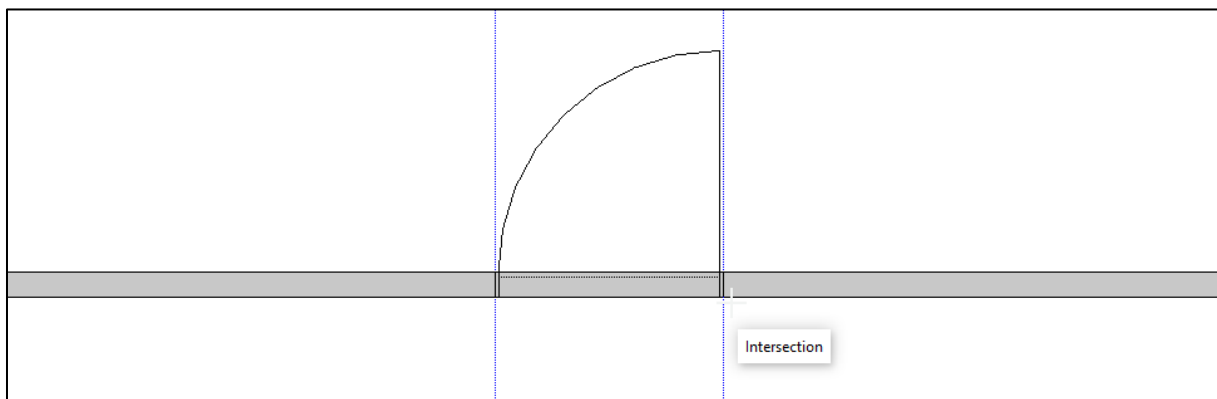


Doors

Go to the “Walls” tab for placing doors in your model. This chapter will explain two types of doors, because these are the most relevant for most buildings. Door number 1 is a “normal” hinged door, door number 2 is a roller door. Click on the black arrow in the bottom right corner of the roller door to select a sectional door.



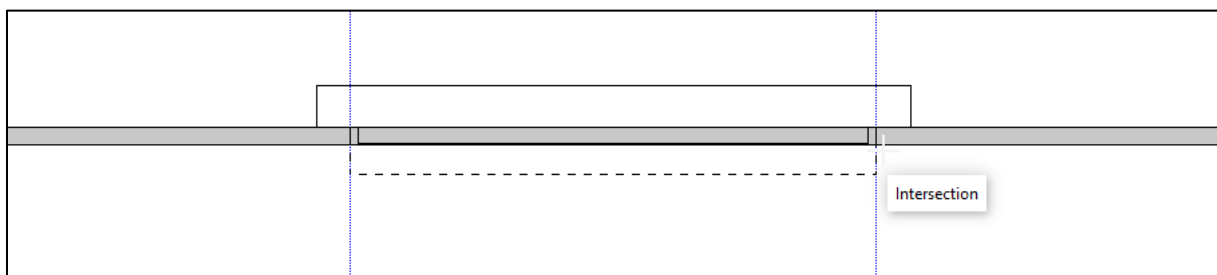
Placing the doors is easiest in the 2D view and the doors can only be placed on walls. When placing door number 1, first click on a point on the wall for placing a side of the door. Then click a second time on a point on the wall to place the other side of the door and set the width.



After placing the door, it is possible to change the dimensions, the direction and the opening angle. A green box will open in the left top corner after selecting the door where these settings can be changed. Make sure that the door is completely closed when rendering, so no outside light comes in from the door. Close the door by changing the Opening angle to zero degrees.

Height	2,115 mm
Width	930 mm
Elevation	0 mm
Flip outside	☐
Hinge left	☒
Opening angle	90°
Door opening arc	Solid
Threshold	☒
Door line width	1
More Properties	

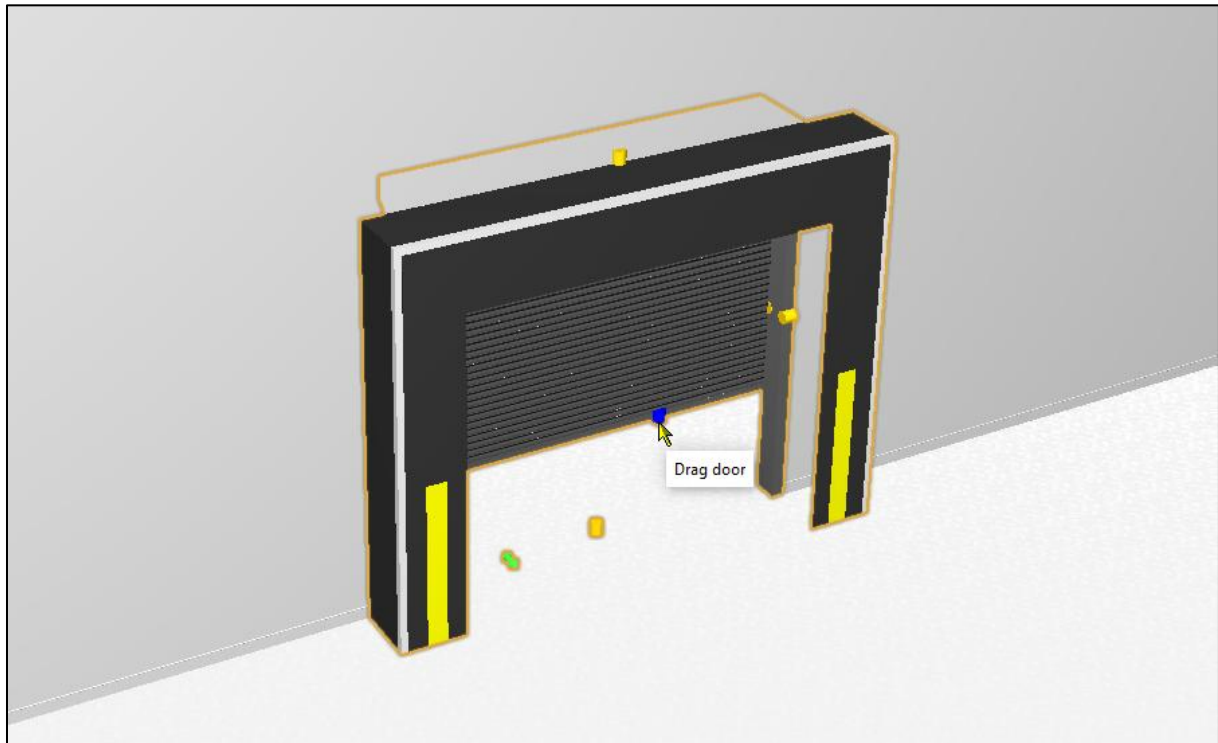
Door number 2 can be placed in the same way as the other door, first click on a point on the wall for placing a side of the door. Then click a second time on a point on the wall to place the other side of the door and set the width.



After placing the door, it is possible to change the dimensions, the direction and add or remove the shelter to the door. A green box will open in the left top corner after selecting the door where these settings can be changed.

Height	3,000 mm
Width	6,270 mm
Elevation	0 mm
Flip outside	☐
Add shelter	☒
Side curtain width	500 mm
Shelter depth	500 mm
Top curtain height	500 mm
Roller type	Standard Polycarbonate Flat
Slat height	50 mm
Guide rail thickness	100 mm
Header box height	350 mm
More Properties	

The opening of the roller door can be changed in the 3D view. First click on the door to select it, now a blue dot becomes visible at the bottom of the door. Click and hold the left mouse button to drag the door to the desired position. Close the door completely for the best rendering results.

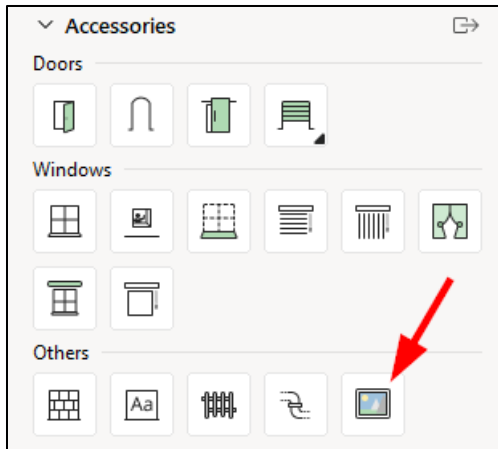


Other accessories

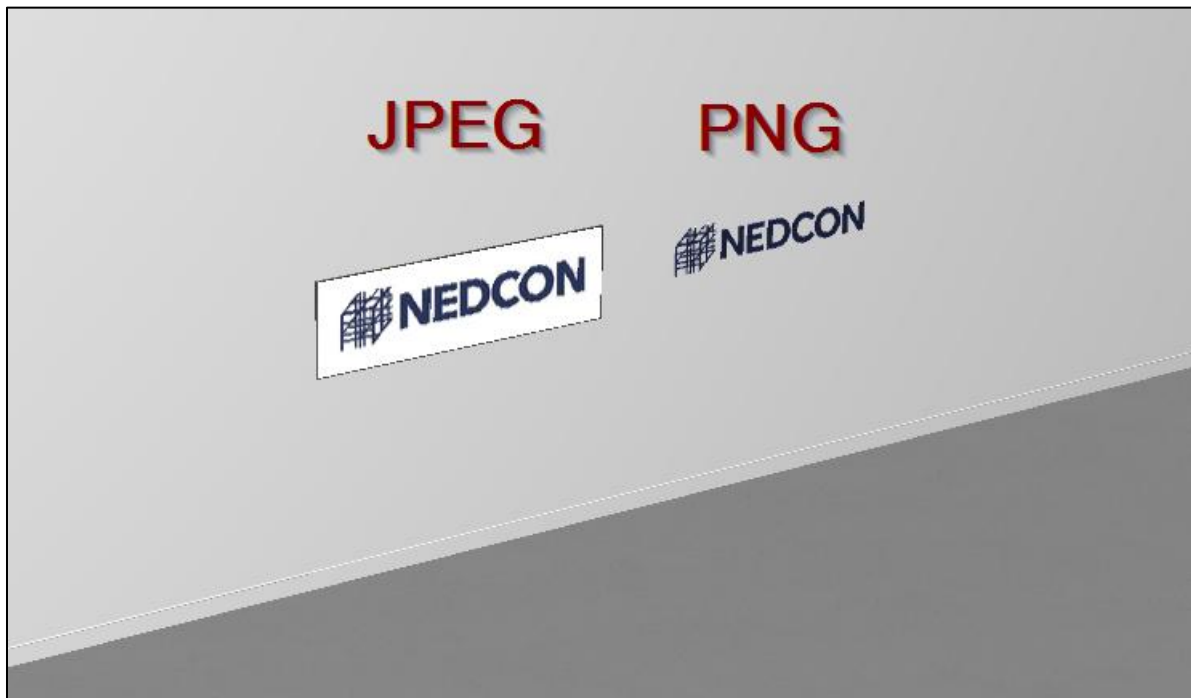
There are a lot of other accessories in CET, more accessories can also be downloaded from the CET Marketplace. One of the most useful accessories for warehouses can be found in the “Industrial equipment” extension, which can be downloaded from CET Marketplace. Here you can find forklifts, pallets, boxes, signs and a lot more. The “Characters” extension can be used to add people to the drawing. These characters can be selected doing different poses, some characters are available operating a forklift or a reach truck.

 Industrial Equipment	 Characters
	Characters Poses
Vehicles	Adam (Height: 1.81m / 5' 11.3")
Industrial Machines	Blake (Height: 1.67m / 5' 5.75")
Pallet	Chris (Height: 1.72m / 5' 7.72")
Trolley	Dee (Height: 1.58m / 5' 2.2")
Industrial Table and Racking	Ed (Height: 1.74m / 5' 8.5")
Industrial Fan	Fran (Height: 1.61m / 5' 3.39")
Hanging Garment	Gary (Height: 1.76m / 5' 9.29")
Box	Holly (Height: 1.63m / 5' 4.17")
Gitterbox	Ian (Height: 1.32m / 4' 4")
Roll cage	Jane (Height: 1.45m / 4' 9.09")
Polybag	Keith (Height: 0.89m / 2' 11.04")
Storages	Lily (Height: 0.89m / 2' 11")
Fire Protection	Misha (Height: 1.75m / 5' 8.89")
Ladders	Nate (Height: 1.85m / 6' 1")
Disposal Area	
Labels and Signs	

It is possible to put your own images on a wall, you can use this to display a logo for example. Go to the “Wall Drawing” tab and click on the “image on wall” icon to begin placing the image. Click on a spot on the wall to set the location of the image. Double click on the image frame to select an image from your files.



Two of the most used image files have a different effect as an image on the wall. JPEG files will always show as a rectangular shape and come standard with a frame. PNG files can have a transparent background and are without a frame, as can be seen in the image below.



To change the setting of the image, first click on the image to select it. Now the green box opens where the image settings can be changed. The numbered settings will be explained below.

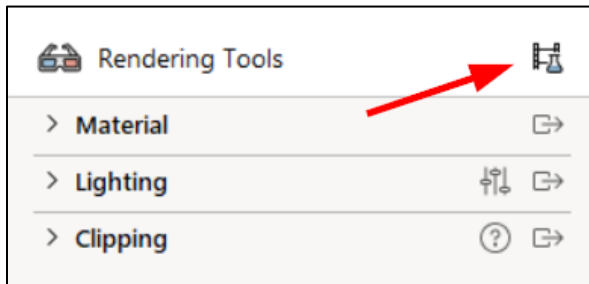
1. Select one of the four predefined image formats or choose "Edit" to select your own format. When "Edit" is chosen, the format will automatically change to the aspect ratio of the selected image. New options become available in the green box.
2. This option is only important when another image size aspect ratio is used as the selected image. It changes how the Image size is filled with the selected picture.
3. Choose if the image is in portrait or landscape orientation. This option is only available when one of the four predefined image formats is chosen.
4. These options are only available when an image with background is chosen. These options can be used to configure the frame around the image.
5. When checking the box, the image will move to the opposite side of the wall.
6. Change the elevation of the image, the elevation is from the floor to the bottom of the image.

The screenshot shows a settings panel for an image. It has a light green background and a grey sidebar on the right with numbered red circles 1 through 6. The settings are as follows:

- Image Size:** 50x70 cm (dropdown menu with an 'Edit' button next to it). Callout 1 points to the 'Edit' button.
- Fill Shape:** Fill (selected), Fit Width, Fit Height. Callout 2 points to the 'Fit Width' button.
- Orientation:** Portrait (selected), Landscape. Callout 3 points to the 'Landscape' button.
- Frame:** None, Thin (selected), Wide, Custom. Callout 4 points to the 'Thin' button.
- Frame Material:** Aluminum (dropdown menu). Callout 4 points to the dropdown arrow.
- Add Matting:** ☐ (checkbox). Callout 5 points to the checkbox.
- Switch Wall Side:** ☐ (checkbox). Callout 5 points to the checkbox.
- Elevation:** 1,000 mm (text input field). Callout 6 points to the input field.

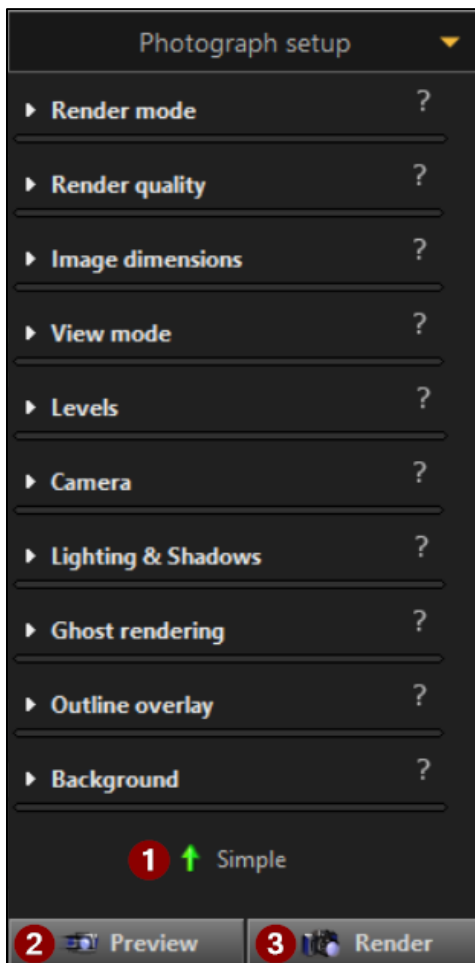
6. Rendering

Advanced rendering is done in Photo Lab. To open Photo Lab, go to “Rendering Tools” and click on the Photo Lab icon. A new window will open with Photo Lab.



On the left side of the screen in Photo Lab, Photograph setup can be found. Here you can set quality size, lighting and other settings for rendering. Some important settings will be explained in chapter [Photograph setup settings](#).

1. Use this button to switch between Advanced and Simple settings within Photograph setup.
2. When the camera is in the correct position, you can use the Preview button to generate a quick render. This render has a much lower resolution than the normal render button.
3. Use this button to create a full resolution render.

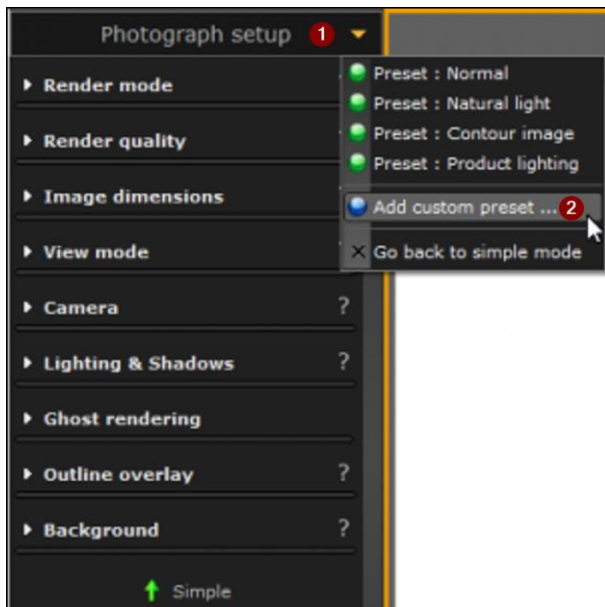


After a render is created, a new render can be made by clicking on the “New” button on the bottom left corner of the screen. Now you are able to move the camera position again to make a new render.

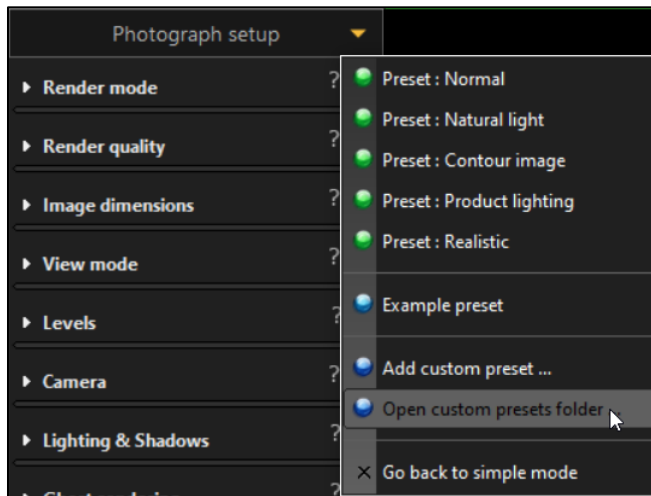


Photograph setup settings

On the left side of the screen in Photo Lab, the Photograph setup can be found. Here the render settings can be changed. Render setting presets can be saved in Photo Lab when in Advanced mode by clicking on the orange arrow next to Photograph setup. A dropdown menu opens where “Add custom preset” can be selected. Give your preset a name and click on OK. Your custom preset can now be found in Simple mode and can be used for future renders.



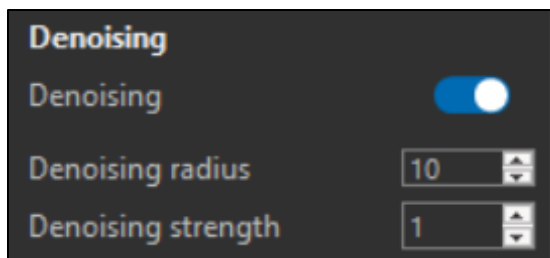
Custom presets can also be shared with others. The folder with the saved presets can be found by clicking the “open custom presets folder” after opening the pulldown menu again. The files in this folder can be shared with others, also preset files from others can be pasted in this folder.



Here are a few important settings which can be found in Advanced mode to enhance the rendering results even more.

Render quality

When there is a lot of noise in the final render, turn the “Denoising” on to reduce the noise. It will result in a longer render time.



Turn the “PBR materials” on to make fully use of the PBR materials in CET, this setting will make the materials in renders more realistic at the expense of a longer render time.

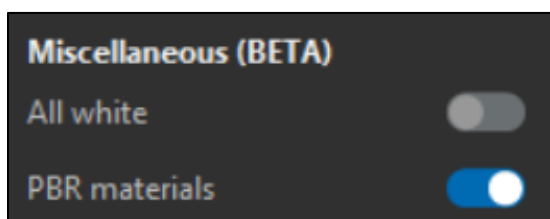
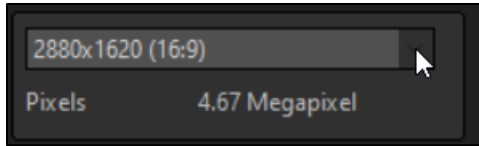


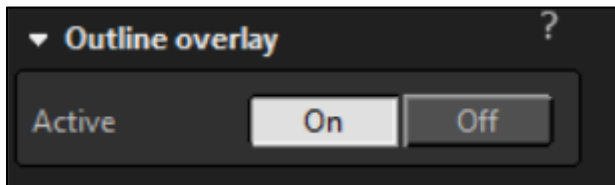
Image dimensions

Click on the arrow next to the current image dimension to change the dimension for rendering. Choose one of the selected dimensions or choose a custom dimension at the bottom of the pulldown menu. The number of pixels which is a result of the chosen dimension has a big influence on the rendering time.



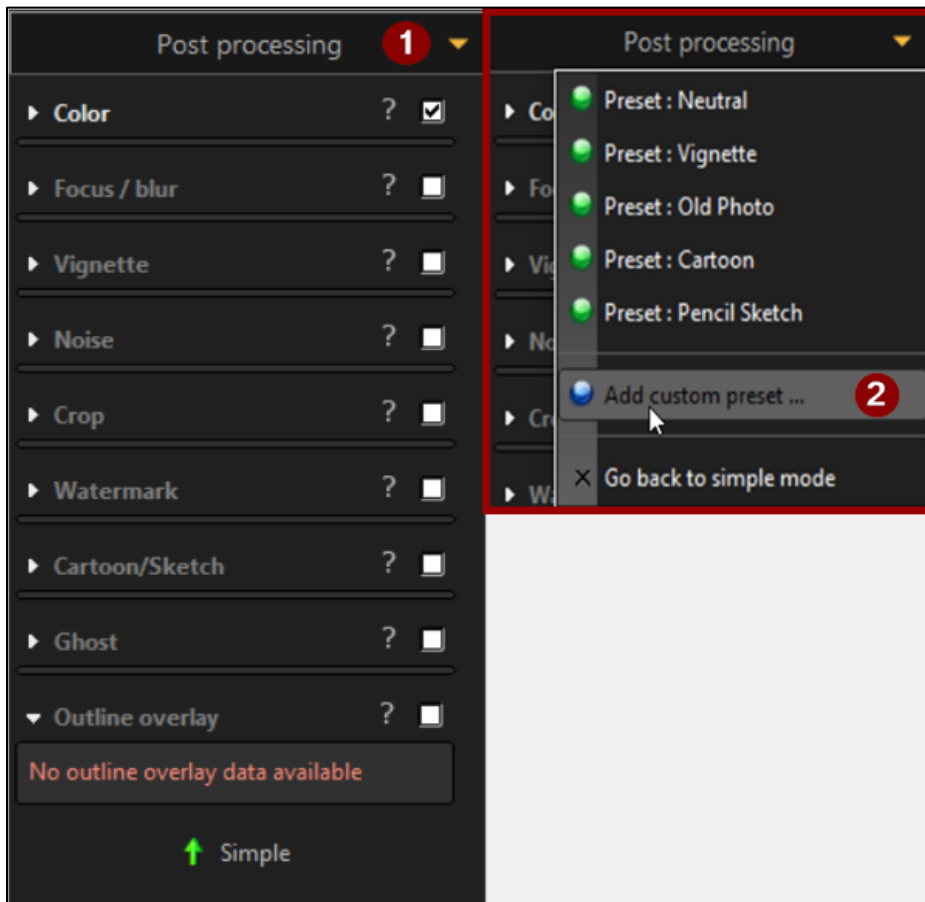
Outline overlay

Turn the “Outline overlay” on to make lines and corners more pronounced in renders. This setting makes it possible to change the opacity and Line thickness during Post processing of the render image. The outline overlay can be turned off in the Post processing when necessary.

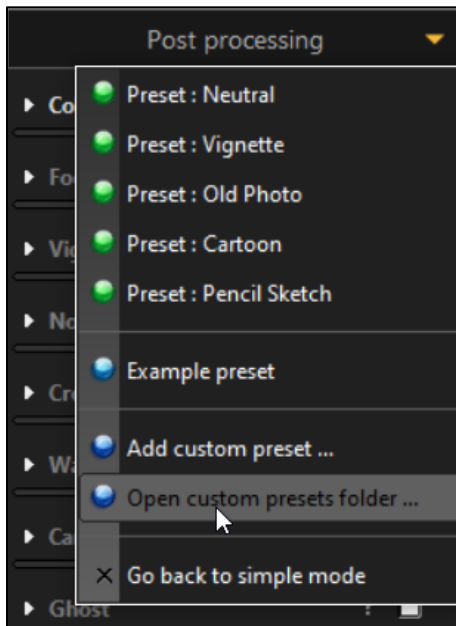


Post processing setting

On the right side of the screen in Photo Lab, the Post processing can be found. Just like the Photograph setup settings, there is a button on the bottom to switch between Simple and Advanced mode. The post processing can only be performed on rendered images. Check the white boxes next to the setting to make the setting active. To finetune the settings open them by clicking on the white arrow next to the setting. Custom post processing presets can be saved by clicking on the orange arrow (1). Now a pulldown menu will open, click on "Add custom preset ..." (2) and enter a name to save the custom preset. Now the custom preset will show up in simple mode for the next renders.



Custom presets can also be shared with others. The folder with the saved presets can be found by clicking the “open custom presets folder” after opening the pulldown menu again. The files in this folder can be shared with others, also preset files from others can be pasted in this folder.

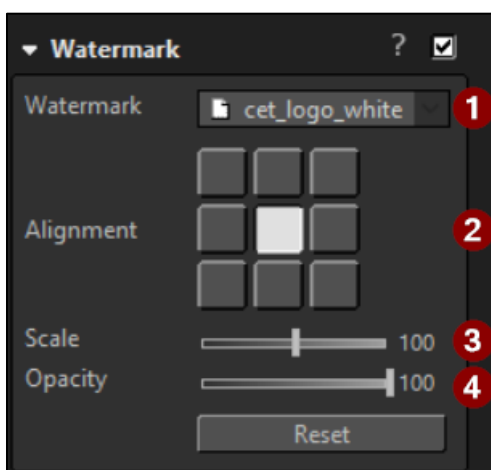


Here are a few important settings which can be found in Advanced mode to enhance the rendering results even more.

Watermark

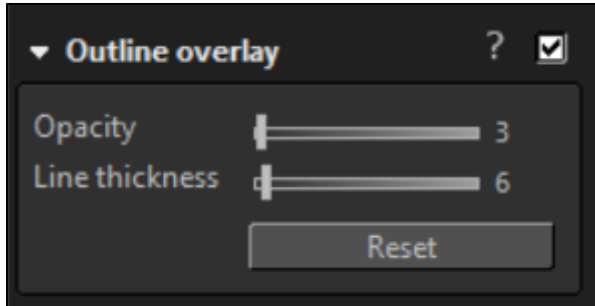
Check the box to add a watermark over the rendered image. The numbered settings are explained below.

1. By clicking the arrow, you can choose or browse for a picture that will be used as a watermark.
2. Select one of the nine squares for positioning the watermark on the render.
3. This slider can be used to scale the size of the watermark.
4. Change the transparency of the watermark with this slider.

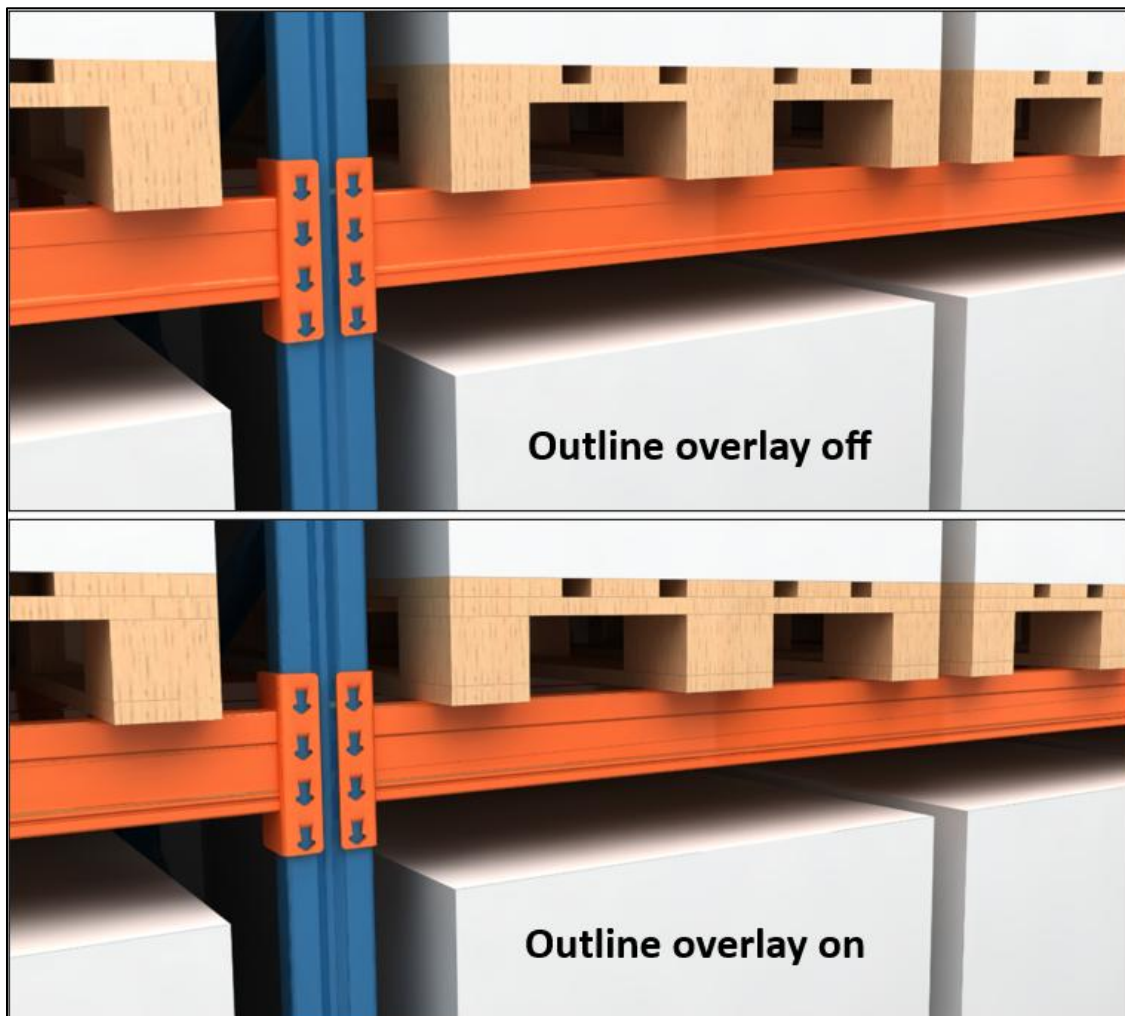


Outline overlay

Outline overlay can only be used in Post processing when the setting was made active in Photograph setup before the image was rendered. This setting makes lines and edges more pronounced by making dark lines visible. The opacity slider changes the transparency of the lines. The Line thickness slider can be used to make the lines appear thicker.



In the picture below the effect of the Outline overlay is displayed. In the picture with the Outline overlay turned on, the opacity is set on value of 10 and the line thickness is set on a value of 11. The effect of the Outline overlay can best be seen on the pallet, where the different layers of wood can be seen.

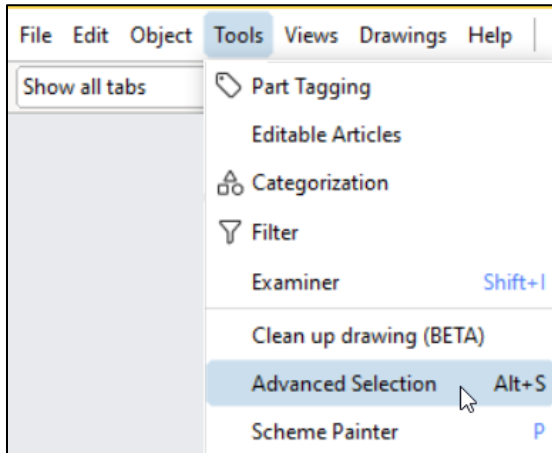


7. Extra tips

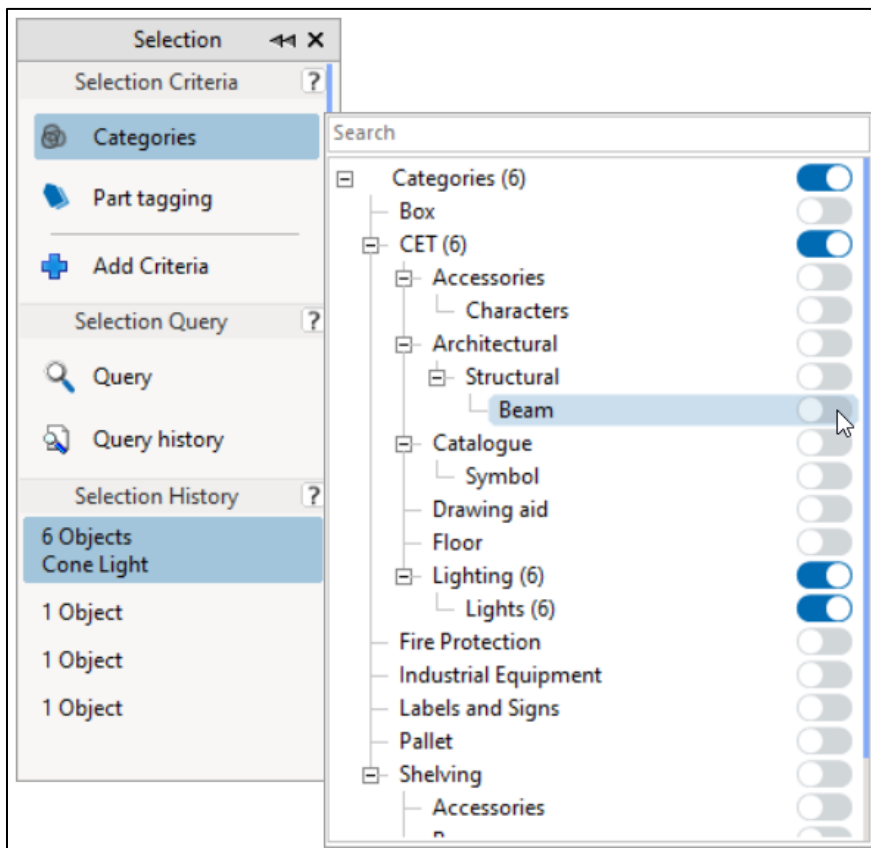
Advanced selection

Advanced selection can be used to select objects in your drawing. It is an alternative way for selecting each object by clicking them. Advanced selection gives more control when selecting a higher number of objects.

Get to the Advanced Selection tool from the Tools drop-down menu.



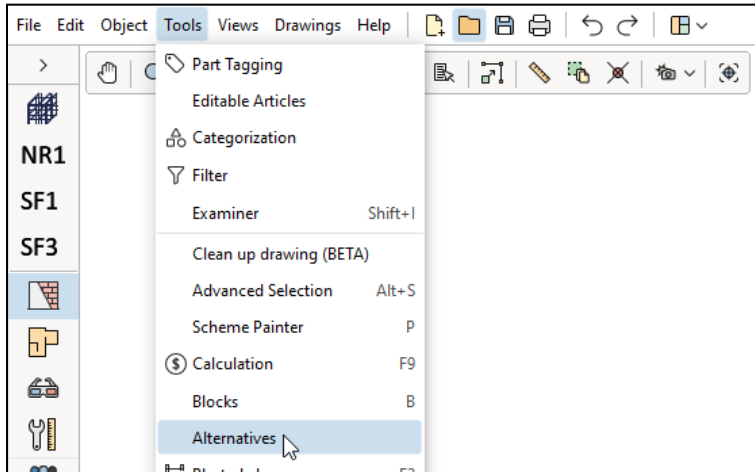
In the Selection Criteria, click on Categories to show all the object categories in your drawing. Turn the toggle switch on for the desired objects to select them. Now the chosen objects will be selected in your drawing.



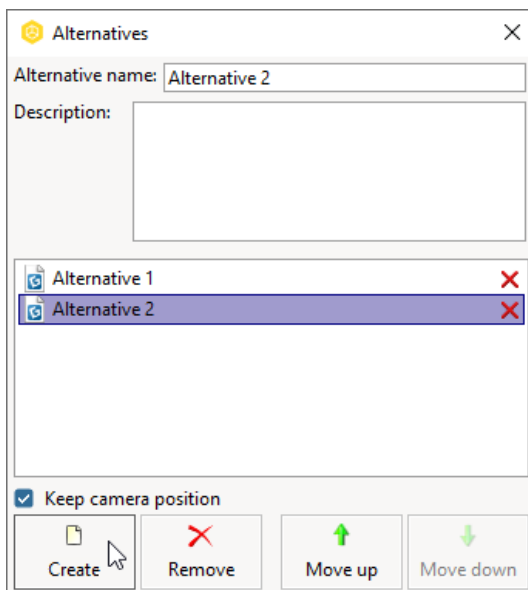
Alternatives

Alternatives can be used to create multiple versions of the same project in the same file. This can for example be used to create a complete version with racking and all kinds of props and an alternative with only the racking. An alternative drawing can be created with the following steps:

1. It is easiest to create the complete drawing first.
2. Select “Alternatives” in the Tool drop-down menu, this will open the “Alternatives” window.



3. Click on “Create” to create an alternative, this will be a copy of the original drawing. Be aware that new alternatives will be a copy from the alternative that is currently active. In the alternative window it is possible to change the name of the alternative and add a description.



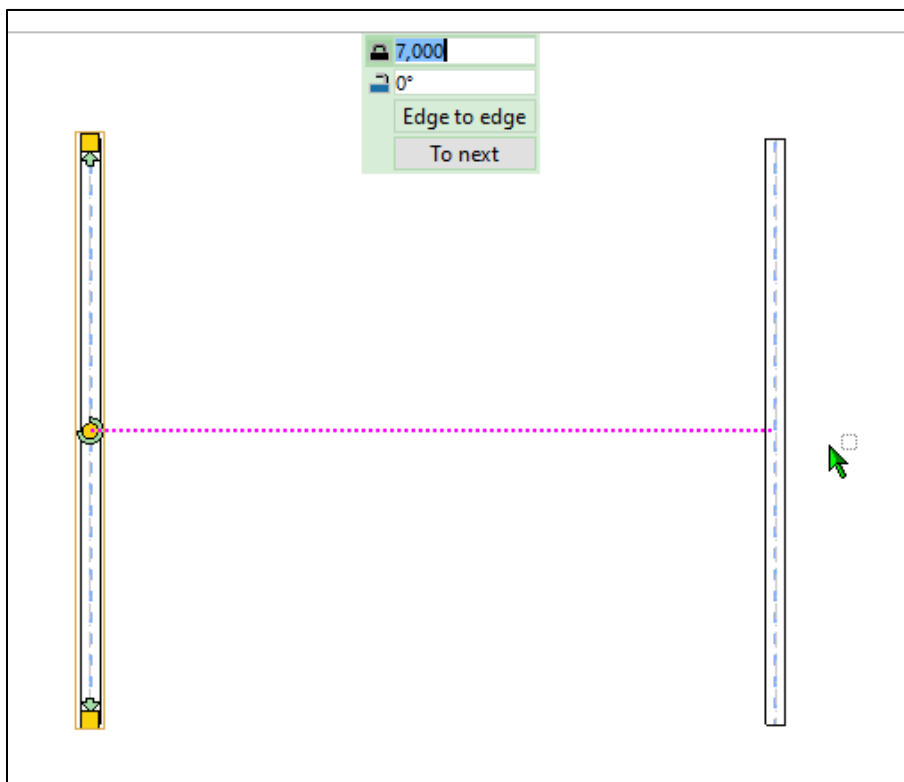
4. You can now edit the alternative, this will only change the alternative that is currently active.
5. Switch between the alternatives using the “Alternative” window.
6. Create as many alternatives as needed.
7. When the “Keep Camera position” checkbox is checked, the camera will hold the position when switching between the alternatives.



Copying objects

Copying objects can be of great benefit when objects need to have the same settings, this is especially beneficial with a larger number of copies. Be aware that not all objects retain the settings from the source object.

First select the source object for the copy, make sure the source object has the correct settings and is in the right position. Use keyboard shortcut Ctrl+C to make a copy from the source object. Use keyboard shortcut Ctrl+V to start pasting the object. The pasted object will follow the mouse cursor, use the arrows on the keyboard to align the pasted object vertically or horizontally with the source object. An extra green box will appear on top of the screen, here the distance between the source and pasted object can be filled in. Click to place the object, the object will remain on your cursor and can be pasted as many times as needed.

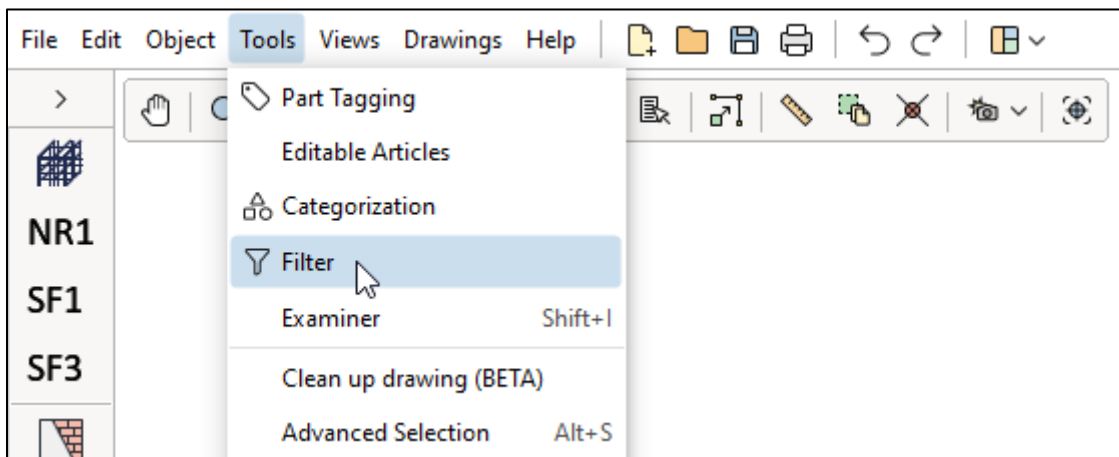


Filters

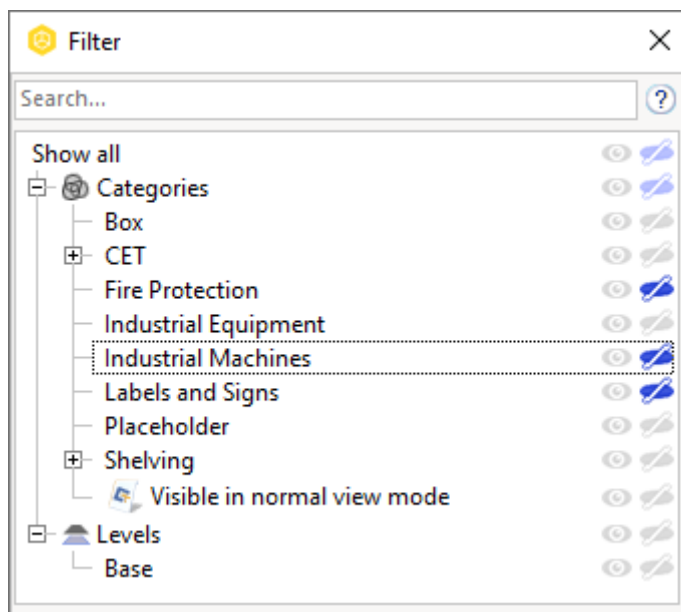
The filter tool can be used to control what is visible in different views. The filter tool works differently for the 2D and 3D views then it does for the Paper view, both methods will be explained separately. Filters applied in the 2D view will not be transferred to the Paper view, nor vice versa.

Filter: 2D and 3D view

The filter tool can be used to control what is visible in the 2D and 3D view. This tool can be used, for example, to hide certain categories during rendering. Open “Filter” in the Tool drop-down menu, this will open the “Filter” window.

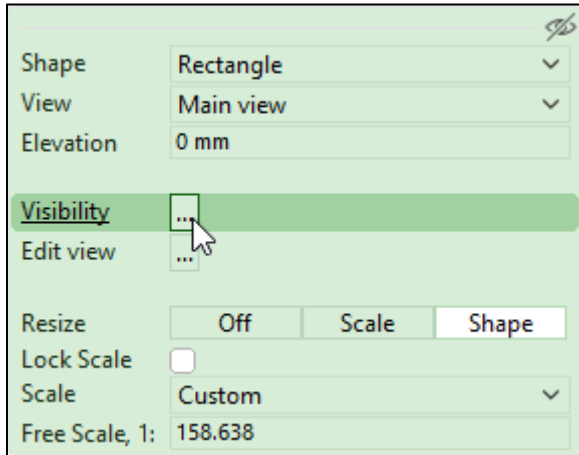


In the “Filter” window, the Categories that are present in the model can be seen. Use the + icon to expand the groups. Click on the grey icon () in the right most column to hide a category. Click on the blue icon (), to unhide a category.

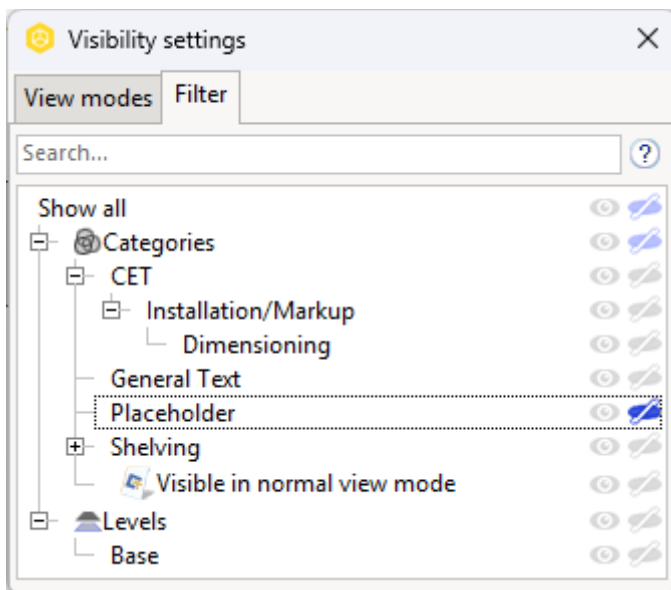


Filter: Paper view

The filter tool in the Paper view is also used to hide categories. First, make sure that a view is added to the Paper view. Select the view, this will make the green box visible. Click on “visibility” in the green box to open the “Visibility settings” window.



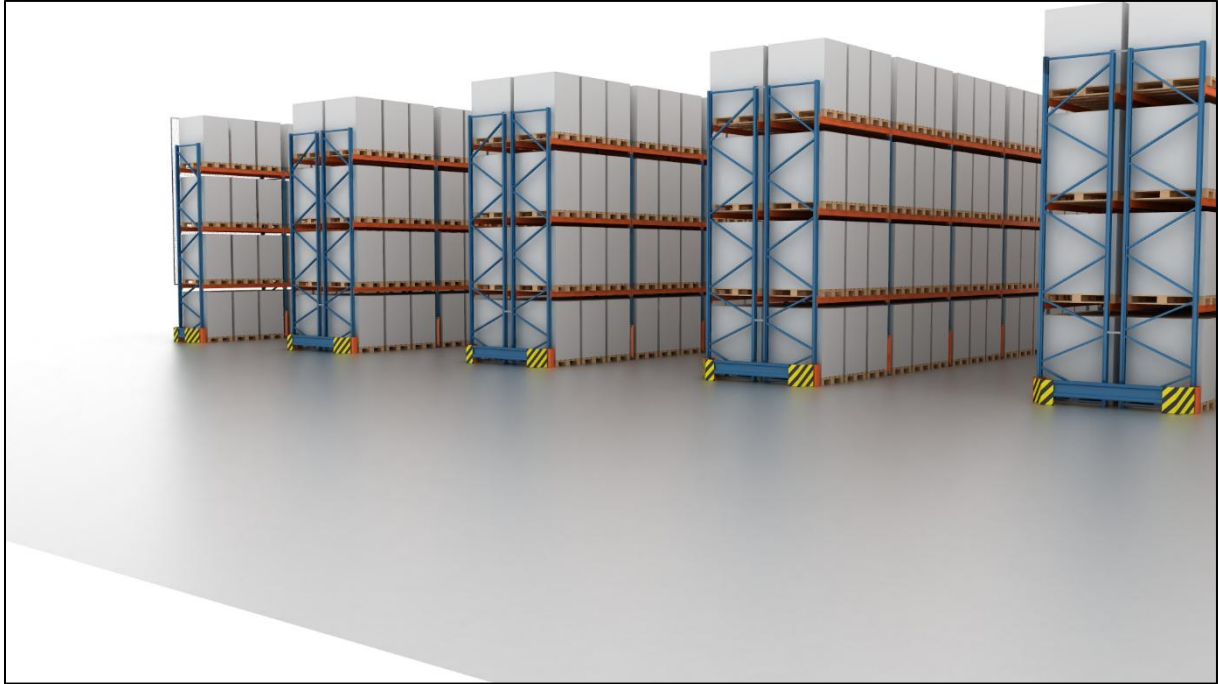
In the “Visibility settings” window, the Categories that are present on the view can be seen. Use the + icon to expand the groups. Click on the grey icon () in the right most column to hide a category. Click on the blue icon (), to unhide a category. This can for example be used to hide standard added dimensions by hiding the “Placeholder” category.



Appendix A: Effect of adding accessories to a project

Here are some examples of how a building and accessories affects the presentation of a racking project. All renders are created from the same camera position and with the same render settings.

In the first example, only the racking is visible.



In the following example, a building is added. The building consists of walls, columns, trusses and indoor lighting.



In the last example, accessories have been added to the building. Such as forklifts, people, a conveyor belt, a workbench and boxes. Some random pallets have been removed from the racking to make it more reflective of reality.



Appendix B: Effect of color temperature

Here are three examples to show which effect color temperature has on renders. All three renders use the same camera position, render settings and light brightness.

Color temperature	Render
Warm (2500 K)	
Neutral (4500 K)	
Cold (6500 K)	